



SEMESTER 1 EXAMINATIONS 2025/2026

MODULE: CSC1046 - OO Analysis & Design

PROGRAMME(S):

CASE	BSc Computer Apps (Sft.Eng.)
COMSCI	BSc Computer Science
ECSA	Study Abroad Program EC
ECSAO	Study Abroad Program EC

STAGE: 1,3

EXAMINER(S):

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TIME ALLOWED: 2 Hours

INSTRUCTIONS: Answer all questions.

PLEASE DO NOT TURN OVER THIS PAGE UNTIL YOU ARE INSTRUCTED TO DO SO.
The use of programmable or text storing calculators is expressly forbidden.

There are no additional requirements for this paper.

QUESTION 1**[TOTAL MARKS: 20]**

Read the Appendix below, describing a "Smart Library Book Kiosk", and complete the following tasks.

Q 1(a)**[15 Marks]**

Write three Use Cases using Cockburn's template for the following scenarios:

- a. A member borrows a book
- b. A member reserves a book that is currently unavailable
- c. A librarian adds a new book to the kiosk

Include appropriate steps, extensions, and preconditions.

Q 1(b)**[5 Marks]**

Draw a Use Case diagram for the "Smart Library Book Kiosk" system. showing relationships (<<include>> or <<extend>> where appropriate) between them.

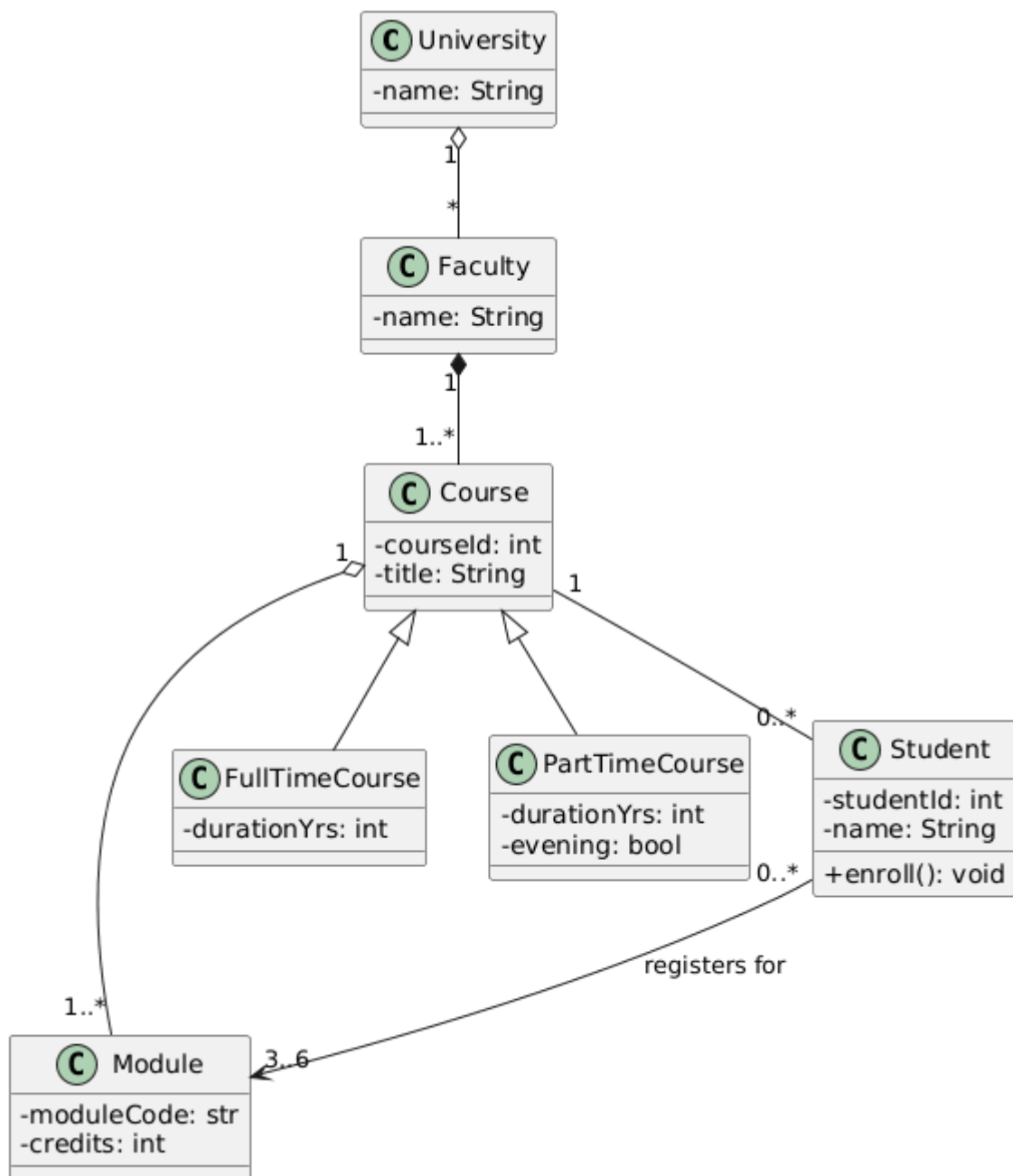
[End of Question 1]

QUESTION 2**[TOTAL MARKS: 20]****Q 2(a)****[10 Marks]**

Explain all of the syntactical elements of the class diagram below. Your explanation must describe the classes, attributes, operations, and the nature of all relationships shown (including association types and multiplicity)

Q 2(b)**[10 Marks]**

Critically evaluate the provided class diagram. Identify one significant weakness in its design with respect to good object-oriented principles (e.g., high cohesion, low coupling, single responsibility). Justify your reasoning and suggest a specific change to the diagram that would improve the model.



Mermaid Description of the class diagram

classDiagram

```
class University {  
  -name: String  
}
```

```
class Faculty {  
  -name: String  
}
```

```
class Course {  
  -courseId: int  
  -title: String  
}
```

```
class FullTimeCourse {  
  -durationYrs: int  
}
```

```
class PartTimeCourse {  
  -durationYrs: int  
  -evening: bool  
}
```

```
class Module {  
  -moduleCode: str  
  -credits: int  
}
```

```
class Student {  
  -studentId: int  
  -name: String  
  +enroll(): void  
}
```

```
University "1" *-- "*" Faculty  
Faculty "1" *-- "1..*" Course  
Course <|-- FullTimeCourse  
Course <|-- PartTimeCourse  
Course "1" -- "1..*" Module  
Course "1" -- "0..*" Student : registers for
```

Textual Description of the class diagram

This diagram shows the relationships between six classes in a university course management system:

University Class (at the top):

Contains one attribute: name (String type)

Has a one-to-many composition relationship with Faculty (indicated by a filled diamond)

Faculty Class:

Contains one attribute: name (String type)

Has a one-to-many composition relationship with Course (indicated by a filled diamond)

Course Class (central to the diagram):

Contains two attributes: courseId (int type) and title (String type)

Serves as a parent class with two child classes connected by inheritance arrows pointing upward to Course

FullTimeCourse Class (inherits from Course):

Contains one attribute: durationYrs (int type)

Connected to Course by a generalization relationship

PartTimeCourse Class (inherits from Course):

Contains two attributes: durationYrs (int type) and evening (bool type)

Connected to Course by a generalization relationship

Module Class (bottom left):

Contains two attributes: moduleCode (string type) and credits (int type)

Has a many-to-many association with Course (multiplicity 1..* on the Course side and 3..6 on the Module side)

Student Class (right side):

Contains two attributes: studentId (int type) and name (String type)

Contains one method: enroll() with void return type

Has a many-to-many association with Course labeled "registers for" (multiplicity 1 on the Course side and 0..* on the Student side)

[End of Question 2]

QUESTION 3**[TOTAL MARKS: 20]****Q 3(a)****[8 Marks]**

Explain the purpose and notation of the following State Machine Diagrams features with examples:

- Substate (Composite State)
- Protocol state
- Join and Fork Nodes
- Junction Node

Q 3(b)**[12 Marks]**

Based on the “Smart Library Book Kiosk” description in the Appendix, draw a State Machine diagram that models the lifecycle of a Kiosk object. Your diagram must correctly use at least one of each of the above features, i.e., Substate, protocol state, join and fork nodes, and junction node, accordingly.

[End of Question 3]

QUESTION 4**[TOTAL MARKS: 20]****Q 4(a)****[8 Marks]**

For a large-scale software system like the “Smart Library Book Kiosk”, high-quality design is crucial. Explain the importance of the following four design principles:

- Robustness
- Flexibility (or Extensibility)
- Correctness
- Usability

Q 4(b)**[12 Marks]**

For each one of the principles mentioned in Q 4(a), describe a specific design decision or technical feature you would implement in the “Smart Library Book Kiosk” system to ensure that principle is met. Justify your answer with example.

[End of Question 4]

QUESTION 5**[TOTAL MARKS: 20]****Q 5(a)****[8 Marks]**

Explain the purpose of a Sequence diagram in modelling object interactions.

Describe the role and UML notation for the following:

- Lifeline
- Synchronous Message
- Create Message
- Interaction Fragment (specifically occurrence and execution)

Q 5(b)**[12 Marks]**

Draw a sequence diagram that illustrates the “A member returns a book” scenario from the Appendix. Your diagram must include an “alt” interaction fragment to correctly model the two possible outcomes: the book becoming “Available” versus becoming “Reserved”. Include an Actor, Boundary, Control, and at least two Entity objects.

[End of Question 5]

APPENDIX

Smart Library Book Kiosk

A new automated book kiosk is being designed for libraries. The kiosk allows members to borrow and return books without staff assistance.

Members can interact with the kiosk via a touchscreen. To begin, a member must scan their library card to authenticate. Once authenticated, they can search for books or browse by category. The system displays the availability and location of each book within the kiosk.

To borrow an available book, the member selects it on the screen, and the system unlocks the transparent door of the corresponding slot, allowing the member to take the book. The system then updates the book's status to 'Borrowed' and links it to the member's account with a due date of three weeks.

If a member wishes to borrow a book that is currently on loan, they can reserve it. The system records the reservation against the member's account.

To return a book, a member scans their card and selects the 'Return' option. A dedicated return slot opens, and the member places the book inside. The kiosk scans the book's internal tag, updates its status, and disassociates it from the member's account. If another member has reserved this book, its status changes to 'Reserved' and it is held for collection. Otherwise, its status becomes 'Available'. A reserved book is held for 48 hours; if not collected, it becomes 'Available' again and the next person on the reservation list (if any) is notified.

Librarians have special access cards to perform maintenance. A librarian can add new books to empty slots or remove a book that is damaged. When a book is removed for repair, its status is updated to 'Under Maintenance'.

[END OF EXAM]